NEXUS AI FINOPS

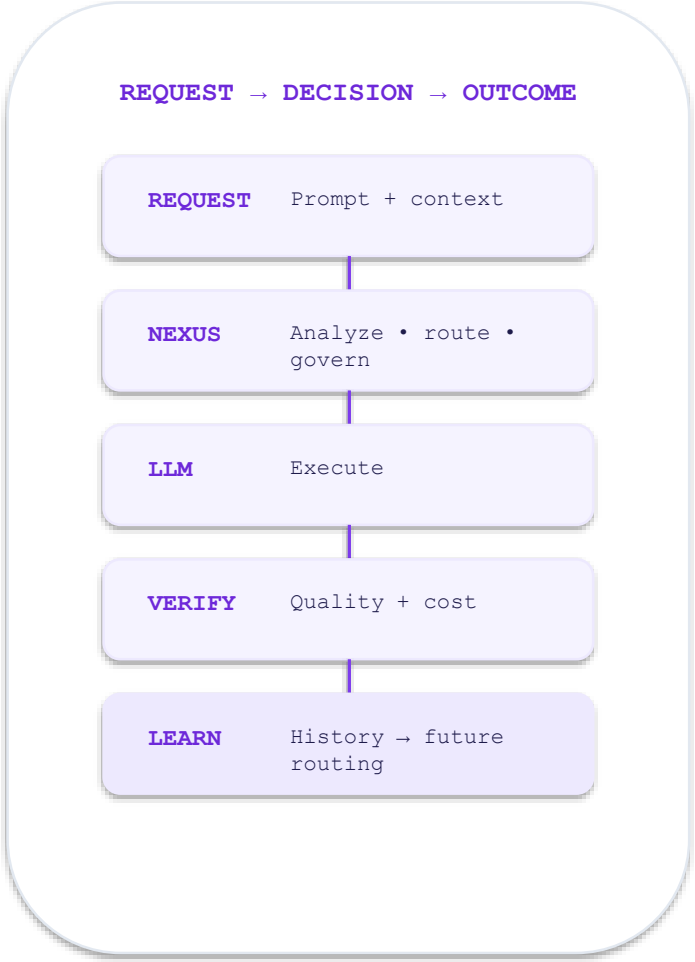
ENTERPRISE AI ECONOMICS

Autonomous AI FinOps & Intelligent Model Routing

A policy-aware AI gateway that automatically selects the lowest-cost capable model, verifies outcome quality, and continuously learns from production behavior.

CORE BUSINESS OUTCOME

Minimize the cost of achieving a successful AI outcome — without silently reducing quality.



Hackathon Solution Provider-agnostic • Real LLM / Demo simulation • Enterprise controls

02

Enterprise Problem & Target State

AI adoption creates a three-way tension: cost, quality and operational constraints.

TODAY

Static or cost-only AI usage

- Expensive models receive simple requests.
- Cheapest models may fail high-risk requests.
- Budgets and SLAs are checked separately.
- Repeated prompts trigger repeated inference.
- Spend dashboards report usage, not business success.

Problem: optimize one variable → break another.

→

TARGET STATE

Closed-loop AI economics

QUALITY	Meet required quality threshold
COST	Choose lowest-cost valid route
LATENCY	Respect SLA / priority
GOVERNANCE	Budget + privacy + policy
RESILIENCE	Fallback when providers fail

North-star KPI: COST PER SUCCESSFUL AI OUTCOME

03

Enterprise Architecture: One Decision Pipeline

NEXUS separates governance, routing, execution and measurement so every request is explainable.

CLIENTS — Web • Mobile • Internal Apps

NEXUS GATEWAY — Request ID • Auth • Rate limits

DECISION LAYER — Analyzer • Cache • Policy • Budget • Router

EXECUTION LAYER — Small • Medium • Advanced • Provider abstraction

CONTROL LOOP — Quality • Fallback • Metering • Historical learning

Every routing decision records: requirements → eligible models → selected model → expected cost → quality evidence → outcome.

04

Explainable Routing: How NEXUS Makes a Decision

The system does not assume that a harder-looking query always requires the most expensive model.

EXAMPLE REQUEST

"Explain recursion with a Java example for a beginner."

SMALL

Quality 88 **₹0.01**

Rejected

Required quality = 91

MEDIUM

Quality 93 **₹0.05**

Selected

Required quality = 91

ADVANCED

Quality 97 **₹0.20**

Valid but costly

Required quality = 91

WHY MEDIUM?

- Small is below the required quality threshold.
- Medium satisfies quality and latency at lower cost than Advanced.
- Historical task performance can increase or decrease a model's effective suitability.

AFTER EXECUTION

Quality = 94% • SUCCESS

If quality < 91% → automatic fallback.

05 Differentiation, FinOps Intelligence & Demo

NEXUS combines capabilities that are usually fragmented across routing, observability and FinOps tools.

01 Cost per Successful Outcome

Measures the economics of success — not merely tokens saved.

02 Task-Specific Historical Learning

Uses observed quality and success by task/model to improve future routing.

03 Quality-Guarded Fallback

Automatically upgrades after a failed quality check; no user intervention.

04 Autonomous FinOps Advisor

Finds traffic-shift opportunities and estimates savings + quality impact.

JUDGE DEMO

1 Simple query → Small

2 Complex query → Advanced

3 Paraphrase → Cache hit

4 Quality fail → Fallback

5 Advisor → Savings action

IMPLEMENTATION

FastAPI • Provider abstraction • MongoDB/SQLite
React • Analytics • Real LLM / Demo mode

NEXUS does not simply route AI traffic. It continuously decides, verifies, measures and learns to reduce the cost of successful AI.